

MECHANICAL ENGINEER

AVATAR SARGAMANTHA NDOEN

MAKER AT HEART.
BUILD. TEST. OPTIMIZE. REPEAT.

CONTACT

Toyota, Japan
arga.ndoen@gmail.com
avatarndoen.github.io

EDUCATION

INSTITUT TEKNOLOGI BANDUNG
Bachelor of Engineering in Mechanical Engineering
2021–2025 | GPA: 3.80/4.00
Undergraduate research: Biomechanics Research Lab

SKILLS

Mechanical Design
Mechanisms, actuators, transmissions, linkages.
Simulation
CFD; static, dynamic, and kinematic analysis.
Control
Motor control, teleoperation.
Computational Design
CAD/CAE automation, ML surrogate modelling, multi-objective optimization

RESEARCH INTERESTS

Medical and Assistive Robotics
Teleoperation
Compliant and Soft Mechanisms
Actuation and Transmissions
Computational Design
Model-Based Control

ABOUT ME

I'm Avatar, a mechanical engineering graduate from Institut Teknologi Bandung. I enjoy designing mechanisms, building robots, and using computational methods to improve their design.

For my undergraduate thesis, I designed a compliant prosthetic wrist and developed a CAD–CAE automation and machine-learning-assisted optimization workflow to personalize its motion and stiffness. My independent projects include compact actuators, a teleoperated robotic arm and wearable controller, a tendon-driven hand, and a continuum robot.

I currently work on automotive sensor integration in Toyota's Body Electronics Design Division in Japan. Outside work, I continue to design, build, and test my own robotics projects.

EXPERIENCE

TOYOTA MOTOR CORPORATION
Mechanical Design Engineer — Body Electronics Design Division
Assignment through CBS Techno | Toyota, Japan | May 2026–Present
Design sensor-mounting components and packaging in CATIA V5; developed a macro to assess sensor coverage, obstruction, and interference.

NAKAYAMA IRON WORKS / ITB deLabo
Product Development & Design Engineer
ITB deLabo (part-time) | Bandung, Indonesia | Jul 2023–Aug 2025
Japan onsite internship| Takeo, Japan | Sep 2024–Feb 2025
Designed heavy-machinery components, robotic mechanisms, and micro-hydropower layouts.

PUBLICATION (UNDERGRADUATE THESIS)

A surrogate-assisted framework for the subject-specific scalable design of compliant prosthetic wrists
Scientific Reports, 16, 32606 (2026) | https://doi.org/10.1038/s41598-026-62052-x
My contributions: mechanical design, CAD–CAE automation, machine learning, optimization, and prototype testing.

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NOTABLE PROJECTS

- Compliant Prosthetic Wrist (Bachelor's Thesis)
- Custom BLDC–Cycloidal Actuators & Teleoperated Robotic Arm
- Beltless Gear CVT
- Tendon-Driven Continuum Robot
- Tendon-Driven Compliant Robotic Hand

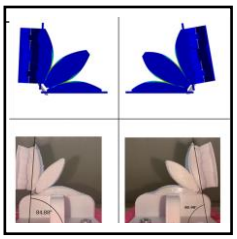
INTERNSHIP / CONTRACT PROJECTS
Nakayama Ironworks, Nusabin

COMPETITION PROJECTS
ABU Robocon, Vorione

ADDITIONAL / FUN PROJECTS
Art, Free Modelling, Machine Learning exploration

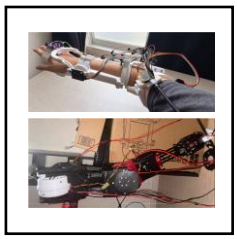
SUMMARY

NOTABLE PROJECTS



Compliant Prosthetic Wrist
(Bachelor's Thesis)
Automated CAD–CAE & Surrogate-Assisted Optimization

Developed a two-DoF compliant wrist and an automated design workflow to optimize geometry for user size, motion, and stiffness targets.



Custom BLDC–Cycloidal Actuators & Teleoperated Robotic Arm
Actuator Design & System Integration

Designed and built custom actuators and a seven-DoF arm, integrating wearable master–slave teleoperation.



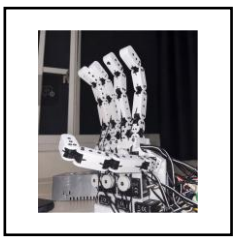
Beltless Gear CVT
Mechanism Design & Computational Optimization

Designed and prototyped a gear-based continuously variable transmission, using computational optimization to refine the mechanism.



Tendon-Driven Continuum Robot
Compliant Mechanisms & Tendon Actuation

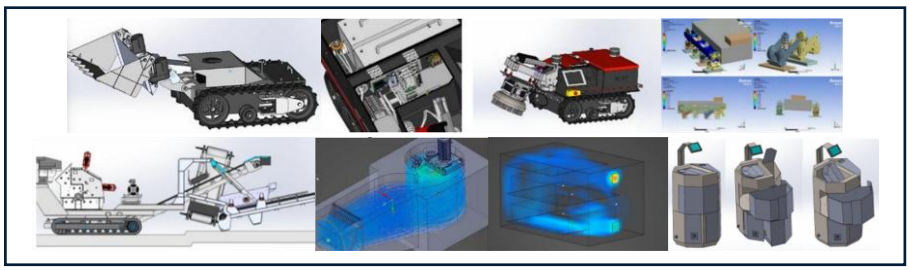
Built a continuum robot with alternating compliant joints and six pulling tendons for bending in two perpendicular planes.



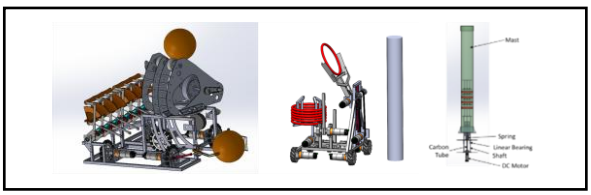
Tendon-Driven Compliant Robotic Hand

Built a 15-DoF robotic hand combining compliant finger joints with tendon-driven actuation.

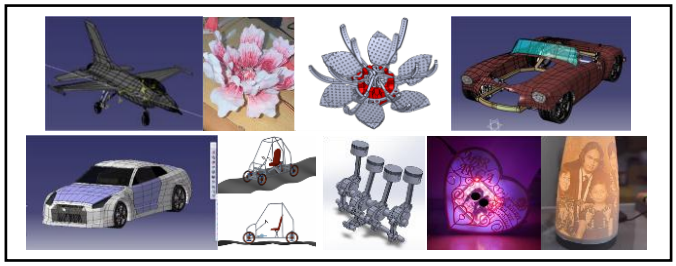
INTERNSHIP / CONTRACT PROJECTS



COMPETITION PROJECTS



ADDITIONAL PROJECTS / FOR FUN



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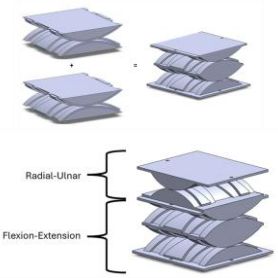
ADDITIONAL / FUN PROJECTS
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NOTABLE PROJECTS

Compliant Prosthetic Wrist

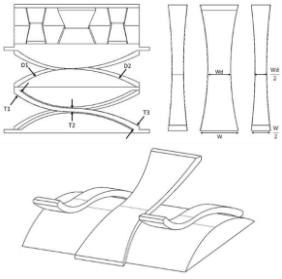
Undergraduate Thesis | Published in Scientific Reports, 2026

Developed a two-DoF compliant prosthetic wrist with an automated CAD–CAE and surrogate-assisted optimization workflow to adapt its geometry to different user sizes while meeting motion and stiffness targets.



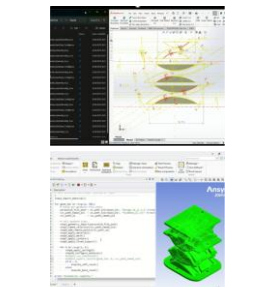
MECHANISM DEVELOPMENT

Asymmetric rolling profiles tune directional motion. Two stacked modules provide flexion–extension; an orthogonal module adds radial–ulnar deviation, creating a two-DoF compliant wrist.



PARAMETRIC DESIGN

Parameterized wrist size, rolling profiles, and flexure dimensions through 21 geometric variables in a master sketch, enabling automated design variation.



AUTOMATED CAD–CAE [VIDEO]

Linked SolidWorks API and ANSYS ACT through geometry and face metadata, automating boundary conditions, contacts, meshing, solving, and response extraction across four motion directions.



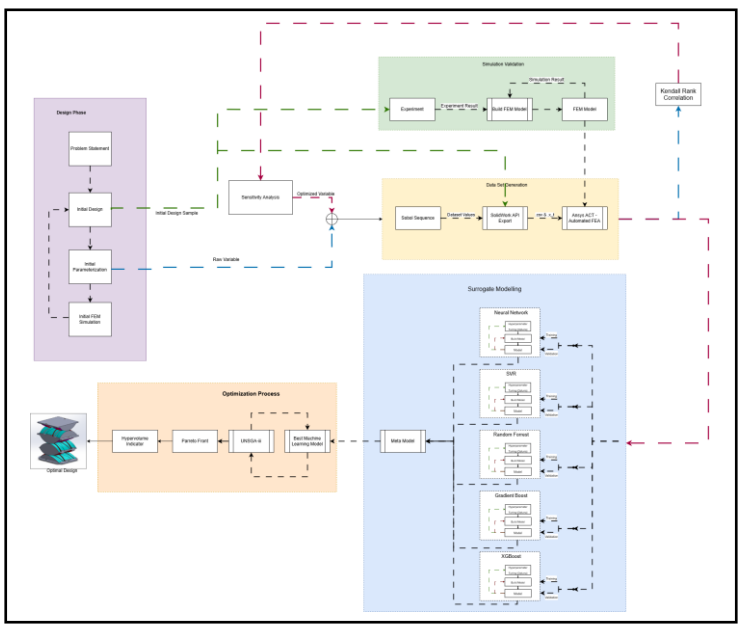
EXPERIMENTAL SETUP

Built a motor-driven tendon-pulling rig with inline force measurement to test the printed prototype's range of motion. prototype's response.



SCALABLE DESIGNS

Generated five optimized wrist sizes with prescribed motion and stiffness targets. Independently re-evaluated each geometry using FEA to check surrogate-based design predictions.



SURROGATE-ASSISTED OPTIMIZATION

Trained machine-learning surrogates on FEA data and integrated them with U-NSGA-III to optimize range of motion and initial/final stiffness under stress constraints.

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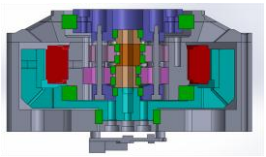
ADDITIONAL / FUN PROJECTS
Art, Free Modelling, Machine Learning exploration

NOTABLE PROJECTS

Custom BLDC–Cycloidal Actuators & Teleoperated Robotic Arm

Designed and built custom BLDC–cycloidal actuators, a wearable encoder-and-IMU master, and a seven-DoF slave arm.

1. CUSTOM CYCLOIDAL ACTUATORS



COMPONENT COLOR KEY
Grey: Housing & Structure · Green: Bearings · Cyan: Rotor · Red: Stator · Purple: Output Assembly · Pink: Cycloidal Discs 1 & 2 · Brown: Eccentric Shaft

ACTUATOR ARCHITECTURE

Integrated a frameless BLDC motor, cycloidal reducer, shafts, bearings, and magnetic encoder within a custom housing.

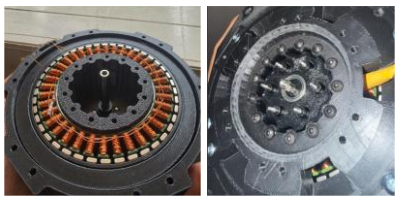
*Thicker structural sections compensate for the limited stiffness of the 3D-printed components.



CONTROL & MOTION TESTING

[VIDEO]

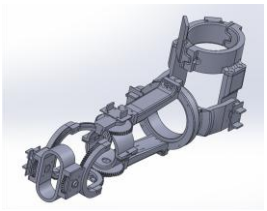
Demonstrated high-speed rotation with minimal visible runout. A preliminary dial-indicator return test measured approximately 0.05 mm deviation at a 200 mm radius; more formal testing remains necessary.



CYCLOIDAL TRANSMISSION

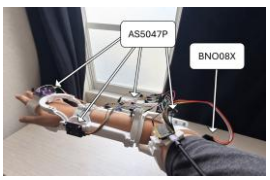
Designed a cycloidal reducer housed inside the rotor for compact packaging, with a 12:1 ratio to balance torque amplification and backdrivability.

2. WEARABLE MASTER SYSTEM



MASTER MECHANISM

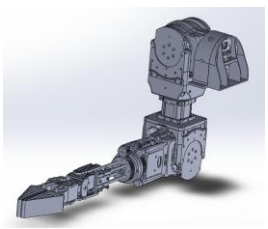
Designed a wearable linkage to capture shoulder, elbow, forearm, and wrist motion for teleoperation, with an open-center, two-DoF wrist allowing hand passage.



ENCODER & IMU SENSING

Four magnetic absolute encoders capture elbow, forearm, and wrist motion; an IMU estimates upper-arm orientation for shoulder tracking.

3. SEVEN-DOF SLAVE ARM



ARM ARCHITECTURE

Built a 3D-printed structure reinforced with carbon-fiber tubes, integrating five custom BLDC actuators and two wrist servos.



SYSTEM INTEGRATION & TESTING

[VIDEO]

Demonstrated wearable master–slave teleoperation, gravity compensation, and motion recording/playback.

AN INTERESTING DESIGN FAILURE



Despite negligible free play, the output exhibited undesired elastic angular deflection under load. Inspection identified output-pin bending as the source of torsional compliance, prompting an upgrade to larger-diameter pins.

WHAT IS NEXT?

Integrate the tendon-driven compliant robotic hand (Project 5) for teleoperated pick-and-place testing.

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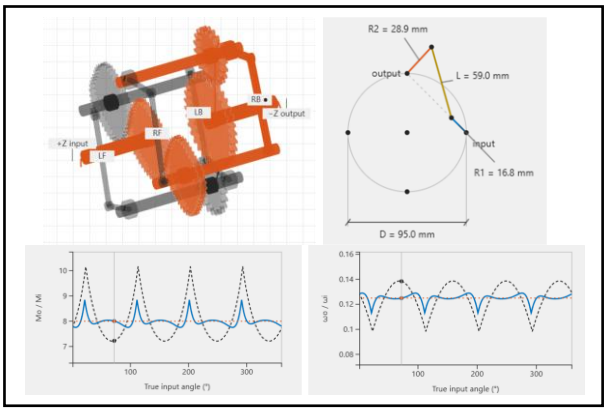
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ADDITIONAL / FUN PROJECTS
Art, Free Modelling, Machine Learning exploration

NOTABLE PROJECTS

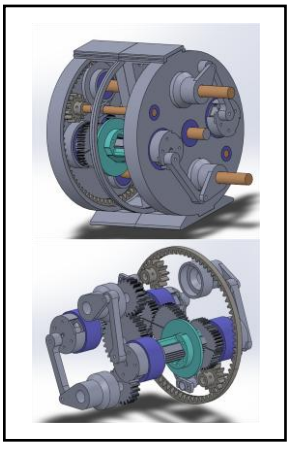
BELTLESS GEAR CVT

Started as a side project to learn machine learning before applying it to my prosthetic-wrist research. Reverse-engineered RatioZero's beltless CVT through mathematical modeling, geometry optimization, SolidWorks Motion gear-engagement checks, and physical prototyping.



PYTHON MODELING AND OPTIMIZATION [VIDEO]

Used a mathematical kinematic model implemented in Python to investigate how linkage dimensions and noncircular gear profiles affected speed and torque ripple, then explored surrogate-assisted optimization to reduce torque and speed ripple.



CAD AND MOTION VALIDATION [VIDEO]

Modeled and developed the CVT assembly in SolidWorks, imported the designed gear profiles, and used contact-driven motion to validate gear meshing.



PHYSICAL CONCEPT PROTOTYPE [VIDEO]

Manufactured and assembled a 3D-printed prototype demonstrating continuous ratio adjustment

*I couldn't test the prototype's performance at the time because I didn't have the funds for a motor and encoder.

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ADDITIONAL / FUN PROJECTS
Art, Free Modelling, Machine Learning exploration

NOTABLE PROJECTS

TENDON-DRIVEN CONTINUUM ROBOT

Designed and built a compliant continuum robot with six pulling tendons and three bending DoF in each of two perpendicular planes.



MECHANICAL DESIGN [VIDEO]

Alternating compliant joints, passive segments, and tendon attachments enable bending in two perpendicular planes, with a compliant gripper for object manipulation.

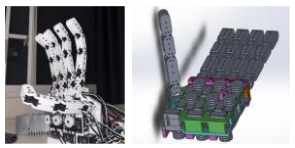


CONTROL & MOTION TESTING [VIDEO]

Demonstrated needle pickup and insertion into a grape, alongside individual bending-DoF actuation to assess motion isolation and parasitic movement.

TENDON-DRIVEN COMPLIANT ROBOTIC HAND

Designed and built a 15-DoF robotic hand combining compliant finger joints with tendon-driven actuation.



MECHANICAL DESIGN

Integrated compliant finger joints and tendon routing into a compact hand structure for articulated finger motion.



CONTROL & MOTION TESTING [VIDEO]

Controlled individual finger movements and demonstrated commanded hand gestures and thumb-to-fingertip contact.

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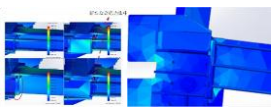
INTERNSHIP / CONTRACT PROJECTS
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ADDITIONAL / FUN PROJECTS
Kinetic Art, Free Modelling, Machine Learning exploration, etc

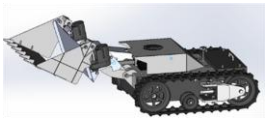
INTERNSHIP / CONTRACT PROJECTS

Mechanical design and simulation contributions to company and startup projects.



NE750J STRUCTURE IMPROVEMENT

Used structural analysis and ML-assisted optimization to improve crawler reinforcement and reduce stress.



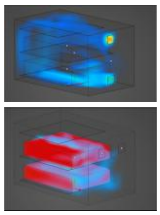
PARALLEL-LINKAGE BUCKET ROBOT

Designed a compact parallel-linkage arm; analyzed motion, structural response, and actuator loading.



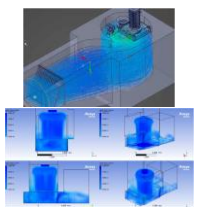
FLOOR POLISH ROBOT

Designed a spring-loaded mechanism, component packaging, sensor mounts, and cable routing. For the use of cement polishing.



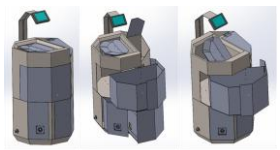
BATTERY THERMAL MANAGEMENT [VIDEO]

Designed vent placement and battery positioning to improve airflow and heat dissipation within the battery pack.



MICRO-HYDRO SYSTEM [VIDEO]

Prepared mechanical layouts and installation drawings; used CFD to verify flow behaviour.



NUSABIN SORTER & COMPOSTER [VIDEO]

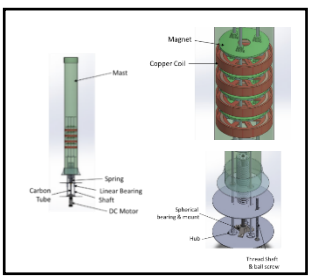
Developed sheet-metal features and mechanisms; supported fabrication planning and sorting–composting system integration.

COMPETITION PROJECTS



ABU ROBOCON / KRAI

Led mechanical design and mechanism development, coordinating integration across a 15-member mechanical, electrical, and control team to build competition robots.



VORIONE BLADELESS WIND TURBINE

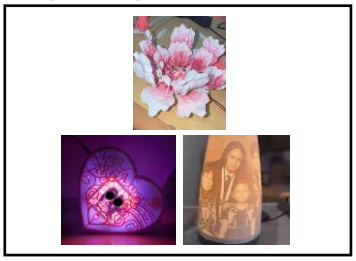
Led mechanical design and vibration analysis for a resonance-based wind-energy prototype, coordinating manufacturing and system integration.

ADDITIONAL / FUN PROJECTS

Explored art projects, free-form CAD modeling, and machine-learning-assisted mechanism design through personal projects.

ART

Kinetic Art [VIDEO], Personal Gift <3, Lithophane Family Portrait [VIDEO]



Free-Form CAD

Surface Modelling [VIDEO], Kinematic-Linkage design [VIDEO]



Computational Design Exploration

ML-Based Kinematic Optimization of Variable Compression Engine [VIDEO] & Suspension Mechanisms

